

PRACTICAL 1

Aim : Connect computer using given topology with wired media. Assume six devices are arranged, if in:

- a) Bus topology
- b) Ring topology
- c) Star topology
- d) Mesh topology

Find out number of cables (links), ports needed in each device and total number of ports needed in entire network for each of above stated topology.

Input :

a) Bus Topology –

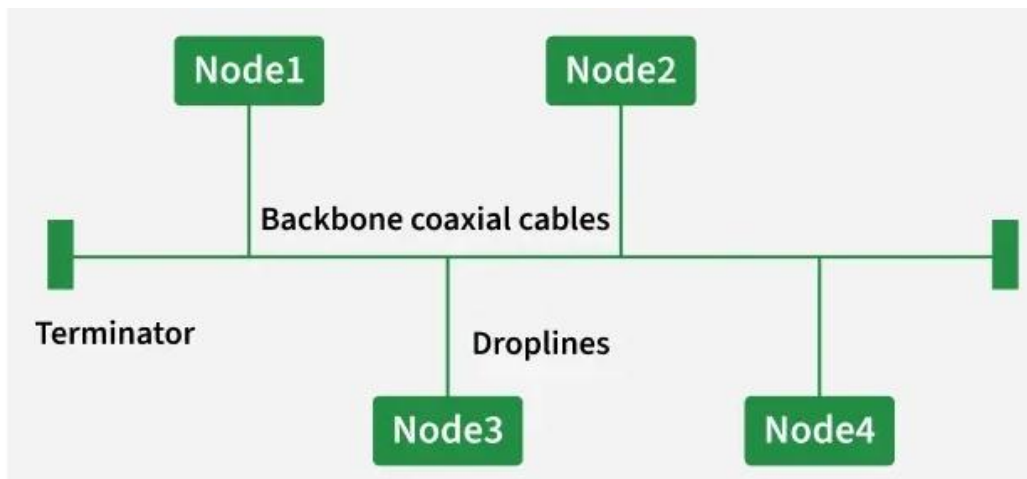
In a bus topology all device are connect to a single central cable/device and the data travel along the bus in both direction, each device only require one port to the bus.

Advantages

- If N devices are connected to each other in a bus topology, then the number of cables required to connect them is 1, known as backbone cable, and N drop lines are required.
- Coaxial or twisted pair cables are mainly used in bus-based networks that support up to 10 Mbps.
- The cost of the cable is less compared to other topologies, but it is used to build small networks.
- Bus topology is familiar technology as installation and troubleshooting techniques are well known.
- CSMA/CD was the only MAC method used in traditional bus Ethernet. Modern switched Ethernet does not use CSMA/CD because full-duplex operation eliminates collisions.

Disadvantages

- A bus topology is quite simpler, but still, it requires a lot of cabling.
- If the common cable fails, then the whole system will crash down.
- If the network traffic is heavy, it increases collisions in the network. To avoid this, various protocols are used in the MAC layer known as Pure Aloha, Slotted Aloha, CSMA/CD, etc.
- Adding new devices to the network would slow down networks.
- Security is very low.



b) Ring Topology –

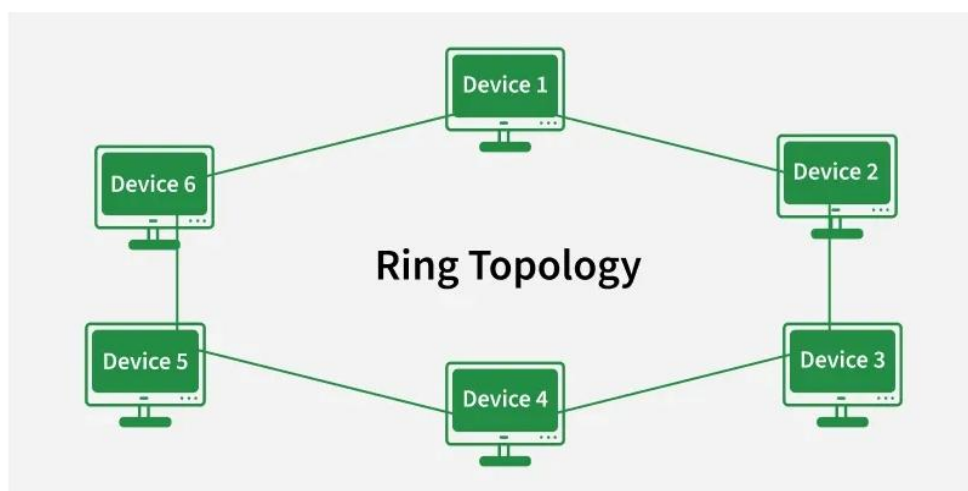
In a ring topology each device is connect to two other device, in a circular faction, so each device has two ports, for incoming data and one port for outgoing data. Data move in unidirectional and bidirectional.

Advantages

- The data transmission is high-speed.
- The possibility of collision is minimum in this type of topology.
- Cheap to install and expand.

Disadvantages

- The failure of a single node in the network can cause the entire network to fail.
- Troubleshooting is difficult in this topology.
- Less secure.



c) Star Topology –

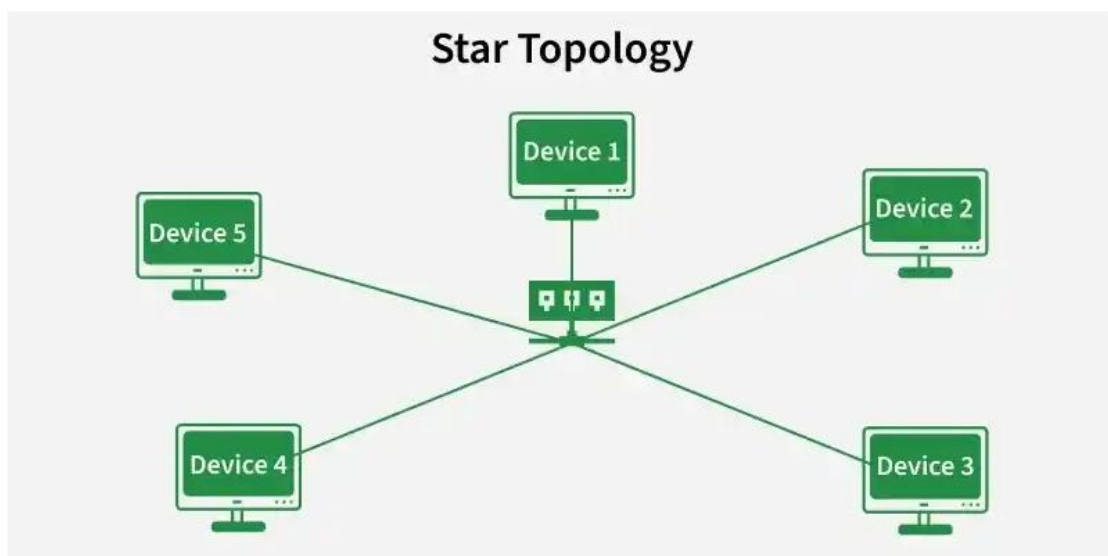
In a star topology each device is connected to a central device (hub/switch) using on cable, the central device manage communication between device.

Advantages

- If N devices are connected to each other in a star topology, then the number of cables required to connect them is N . So, it is easy to set up.
- Each device requires only 1 port i.e. to connect to the hub, therefore the total number of ports required is N .
- It is Robust. If one link fails only that link will affect and not other than that.
- Easy to fault identification and fault isolation.
- Star topology is cost-effective as it uses inexpensive coaxial cable.

Disadvantages

- If the concentrator (hub) on which the whole topology relies fails, the whole system will crash down.
- The cost of installation is high.
- Performance is based on the single concentrator i.e. hub.



d) Mesh Topology –

In a mesh topology every device is directly connect to every other device, in this case the number of connection increase significantly with number of device.

Advantages

- Communication is very fast between the nodes.
- Mesh Topology is robust.
- The fault is diagnosed easily. Data is reliable because data is transferred among the devices through dedicated channels or links.
- Provides security and privacy.

Disadvantages

- Installation and configuration are difficult.
- The cost of cables is high as bulk wiring is required, hence suitable for less number of devices.
- The cost of maintenance is high.

